

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Second-Hand Product Escrow Marketplace #39

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements for a Second-Hand Product Escrow Marketplace. The system is intended to support secure peer-to-peer transactions in which buyer payment is held in escrow until the product has been inspected and the buyer has approved it, while also providing a controlled dispute process.

1.2 Scope

The system covers the escrow transaction lifecycle from payment funding through inspection, buyer verification, fund release, and dispute handling. It also includes security controls for sensitive escrow and payment information.

1.3 System Overview

The marketplace connects buyers and sellers and introduces an inspection and escrow workflow. A buyer funds an order, the payment is locked in escrow, an inspection agent records the condition of the product, and the buyer approves or rejects the item. Funds are released according to the approval/timeout rules, while disputes can freeze the escrow process for resolution.

2. Stakeholders and Actors

- Buyer – purchases a second-hand product, reviews inspection results, approves/rejects the item, and can raise a dispute.
- Seller – sells the product and participates in the order/escrow lifecycle.
- Inspection Agent – inspects the product and submits a pass/fail report with photographic evidence.
- Administrator – manages and resolves disputes and supports marketplace administration.
- Payment/Escrow Service – securely holds and releases buyer funds according to the defined escrow rules.

3. Functional Requirements

ID	Requirement	Description	Priority
FR-001	Fund order into escrow	The system shall lock the buyer's payment in escrow and move the order to the Under Inspection state.	High
FR-002	Submit inspection report	The inspection agent shall submit a pass/fail	High

		inspection report with photographic evidence.	
FR-003	Buyer approval/rejection	The buyer shall be able to approve or reject the inspected item within 48 hours.	High
FR-004	Release escrow funds	The system shall release escrow funds after buyer approval or automatically after 48 hours when the inspection has passed.	High
FR-005	Raise dispute	The buyer shall be able to raise a dispute when there is a problem with the transaction or inspection outcome.	High

4. Non-Functional Requirements

ID	Requirement	Description
NFR-001	Dispute safety	When a dispute is raised, escrow funds and withdrawals shall be frozen until the dispute is resolved.
NFR-002	Security	Payment and escrow-related sensitive data shall be protected using encryption.

5. Use Cases

Use Case	Name	Summary
UC-01	Fund Order into Escrow	Buyer funds an order and the system locks the payment in escrow.
UC-02	Submit Inspection Report	Inspection agent submits pass/fail results and photographic evidence.
UC-03	Review and Approve/Reject Item	Buyer reviews the inspection result and approves or rejects the item within the allowed period.
UC-04	Release Escrow Funds	System releases funds following buyer approval or the defined automatic-release condition.
UC-05	Raise Dispute	Buyer raises a dispute; the escrow process is frozen for resolution.

6. Main Use-Case Flow

- Buyer places an order and funds it.
- System locks the payment in escrow and marks the order Under Inspection.
- Inspection agent inspects the product and submits a pass/fail report with photographic evidence.

- Buyer reviews the inspection report and approves or rejects the item within 48 hours.
- If the item is approved, escrow funds are released.
- If the inspection passed and the buyer does not respond within 48 hours, the system automatically releases the funds.
- If a dispute is raised, the escrow funds and withdrawals are frozen until resolution.

7. Exception Flows

- Payment/funding failure: the order shall not proceed to inspection until valid escrow funding is completed.
- Inspection failure: the failed inspection result shall be recorded and the transaction shall follow the defined rejection/refund or dispute path.
- Buyer rejection: the order shall not be treated as approved and shall follow the applicable return/refund/dispute process.
- Buyer timeout: after the specified 48-hour period, automatic release occurs only when the inspection has passed.
- Dispute raised: escrow funds and withdrawals remain frozen until the dispute is resolved.

8. Acceptance Criteria Summary

- Escrow payment is locked when the order is funded.
- Inspection report contains a pass/fail result and photographic evidence.
- Buyer can approve/reject within 48 hours.
- Funds are released after approval or after the defined successful-inspection timeout.
- Dispute action freezes escrow funds and withdrawals.
- Sensitive payment/escrow information is encrypted.

9. Traceability

The requirements in this SRS correspond to the functional and non-functional requirements defined for the project and are intended to remain traceable to the project's use cases, diagrams, exception flows, work breakdown, and implementation tasks.